INTRODUCTION

Previous lessons examined the payoff and profit profiles of call options and put options, the upper and lower bounds of an option's value, and the factors impacting option values. In doing so, we contrasted the asymmetry of one-sided option payoffs with the linear or symmetric payoff of forwards and underlying assets.

We now extend this analysis further to show that there are ways to *combine* options to have an equivalent payoff to that of the underlying and a risk-free asset as well as a forward commitment. In the first lesson, we demonstrate that the value of a European call may be used to derive the value of a European put option with the same underlying details, and vice versa, under a no-arbitrage condition referred to as put–call parity. In the second lesson, we show how this may be extended to forward commitments and how the put–call parity relationship may be applied to option and other investment strategies. We will focus on European options on underlying assets with no income or benefit.

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Category
Option Replication Using Put-Call Parity

Learning Module Overview

- Put–call parity establishes a relationship that allows the price of a call option to be derived from the price of a put option with the same underlying details and vice versa.
- Put–call parity holds for European options with the same exercise price and expiration date, representing a no-arbitrage relationship between put option, call option, underlying asset, and risk-free asset prices.
- If put–call parity does not hold, then riskless arbitrage profit opportunities may be available to investors.
- Put–call forward parity extends the put–call parity relationship to forward contracts given the equivalence of an underlying asset position and a long forward contract plus a risk-free bond.
- Under put–call forward parity, we may demonstrate that a purchased put option and a sold call option are equivalent to a long risk-free bond and short forward position, and a sold put and purchased call are equivalent to a long forward and short risk-free bond.
- Put–call parity may be applied beyond option-based strategies in finance—for example, to demonstrate that equity holders have a position equivalent to a purchased call option on the value of the firm with unlimited upside, while debtholders have a sold put option position on firm value with limited upside.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. Which of the following statements correctly describes the equivalent to a long position in an underlying according to put–call parity?

A. Long a put option on the underlying, short a call option on the underlying, and long a risk-free bond

C. Short a put option on the underlying, long a call option on the underlying, and short a risk-free bond

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Solution:

B is correct. Put–call parity demonstrates that a long underlying position is equivalent to a sold put option, a purchased call option, and a long risk-free bond. This is shown in the following equation:

Put–call parity: $S_0 + p_0 = c_0 + X(1 + r)^{-T}$ implies $S_0 = -p_0 + c_0 + X(1 + r)^{-T}$.

2. Identify which of the following positions has the same no-arbitrage value as which portfolio under put–call parity:

1. Long call option (c_0)	A. Long underlying, short risk-free bond, and short call option
2. Short risk-free bond ($-X(1 + r)^{-T}$)	B. Long underlying, long put option, and short risk-free bond
3. Short put option ($-p_0$)	C. Short underlying, long call option, and short put option

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Solution:

Recall that the put–call parity relationship may be expressed as

$$S_0 + p_0 = c_0 + X(1 + r)^{-T}.$$

1. B is correct. A long call option position is the no-arbitrage equivalent of a long underlying position, a long put option, and a short risk-free bond position.

2. C is correct. A short risk-free bond position is equivalent to a short underlying position, a long call option, and a short put option.

3. A is correct. A short put option is equivalent to a long underlying position, a short risk-free bond, and a short call option.

3. Which of the following statements correctly describes a synthetic protective put position according to put–call forward parity?

A. A long forward contract on the underlying, a long put option on the underlying, and long a risk-free bond
B. A short forward contract on the underlying, a long put option on the underlying, and short a risk-free bond
C. A short forward contract on the underlying, a short put option on the underlying, and short a risk-free bond

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Solution:

A is correct. The formula for put–call forward parity is as follows:

$$F_0(T)(1 + r)^{-T} + p_0 = c_0 + X(1 + r)^{-T}.$$

4. Which of the following statements best describes a shareholder's claim in terms of an option payoff?

A. Shareholder payoff resembles the payoff of a put option on firm value.
B. Shareholder payoff resembles the payoff of a covered call option on firm value.
C. Shareholder payoff resembles the payoff of a call option on firm value.

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RN Question 3 solution is incorrect

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